

PES UNIVERSITY - DEPARTMENT OF CSE

Software Engineering Lab - Lab 3: Component Modelling & Architecture Selection

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1. Objective & Learning Outcomes

Evaluate architectural styles, select the optimal pattern for Feature Flag Manager, and construct formal UML 2.0 Component Diagrams with ball-and-socket notation.

2. Architectural Style Comparison

- Layered Architecture:** Monolithic; database bottleneck under 50k req/s [Poor Fit].
- Client-Server:** Centralized server exhaustion under 100k+ SDK connections [Poor Fit].
- Event-Driven Microservices:** Independent autoscaling & sub-1ms edge caching [Selected: Best Fit].

3. Architecture Selection Statement

4. Identified Components & Interfaces

- 1. Admin Dashboard Component - React SPA Web Console [IFlagAdminService]
- 2. API Gateway & Auth Guard Component - Reverse proxy & RBAC [mTLS]
- 3. Flag & Rollout Manager - Environment overrides & Kill-Switch [IRolloutConfig]
- 4. Rule Evaluation Engine - MurmurHash3 consistent hashing [IRuleEvaluation]
- 5. Streaming Distribution Service - Real-time SSE/WebSocket hub (<2s delta sync)
- 6. Audit & Incident Dispatcher - Immutable audit ledger & PagerDuty alerts
- 7. Persistence Tier - PostgreSQL Relational DB + Redis Distributed Cache
- 8. Client SDK & In-Memory Evaluator - Sub-1ms local cached evaluation

Key Interfaces: REST/HTTPS, gRPC/mTLS, SSE/WebSockets, Redis RESP, SQL/TCP

5. Written Architectural Justification (1-Page Formal)

A. Architectural Choice:

We selected an Event-Driven Microservices Architecture with Edge-Cached SDK Evaluation.

B. Two Specific Scenario-Related Reasons:

1. Asymmetric Workload & Independent Autoscaling:

Administrative mutations occur at low frequency (<10 req/min), whereas SDK evaluation traffic occurs at immense scale (>100,000 req/sec). Microservices enable independent scaling of Streaming Services without over-provisioning database-backed CRUD services.

2. Fault Isolation & Blast Radius Containment:

Under high stress or database downtime, the critical flag evaluation pipeline remains fully functional because SDKs evaluate against local memory and Redis caches.

C. Security Advantage:

The API Gateway enforces Role-Based Access Control (RBAC) and MFA on write paths. Internal microservices communicate over mutual TLS (mTLS) with identity tokens, isolating sensitive production override rules from publicly accessible SDK endpoints.

D. Performance Benefit:

Using local in-memory SDK cached evaluation updated via Server-Sent Events (SSE), evaluations execute in <1ms without network overhead, and global rollout changes propagate across distributed fleets in <2 seconds without database bottlenecks.

6. UML Component Diagram & Artifacts

The vector UML 2.0 Component Diagram with ball-and-socket assembly connectors is available in the repository at [assets/component_diagram.svg](#) & [component_diagram.puml](#).